LHCb XY surface setup: lhcb_lower_up_gdml_ft_8_8_10

Selected reusable detector planes for transport and acceptance masks.

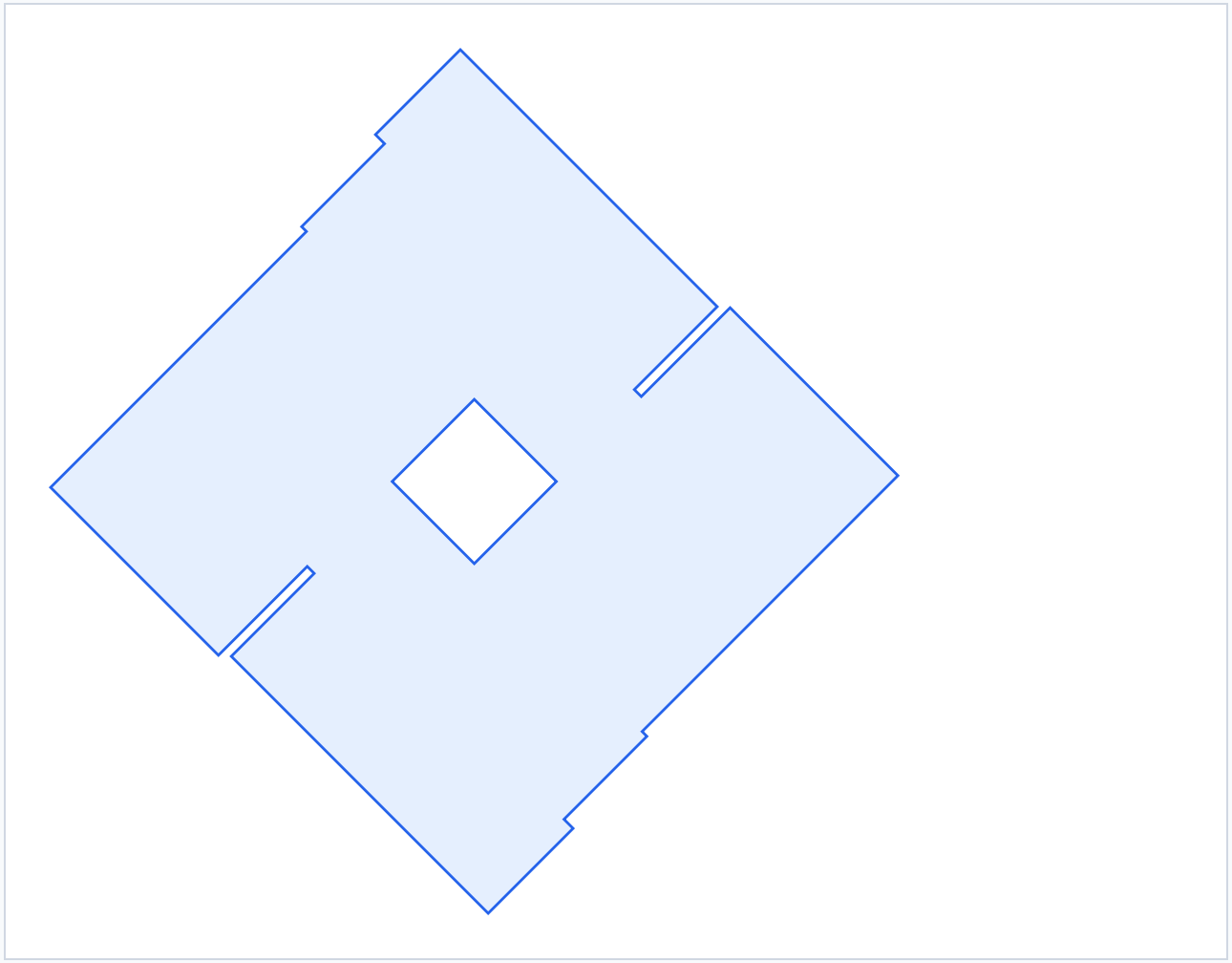
detector	surfaces	fired layers	z range [mm]	XY envelope [mm]	description
TV	50	50	-287.5 to 557.6	90.3 x 92.0	VELO/TV layers formed by unioning the nearby-z sensor faces in each layer.
RICH1	2	2	795.0 to 2237.2	1384.1 x 1142.5	Projective RICH1 radiator entry/exit surfaces from angular acceptance.
UP	16	4	2314.0 to 2656.0	1393.7 x 1279.8	Upstream Tracker surfaces; choose nominal GDML or Yisheng-inspired border crops.
MP	5	5	7665.5 to 9132.5	3288.4 x 672.6	Five Mighty Pixel layers, each formed from the union of its extracted tiles.
FT	12	12	7902.7 to 9384.7	5652.3 x 4798.7	Mighty Tracker X,U,V,X stations; choose 8/10/12 module widths per T station.
RICH2	2	2	9487.5 to 11839.2	2855.1 x 2375.8	Projective RICH2 radiator entry/exit surfaces from angular acceptance.
PicoCal	2	2	12565.9 to 12601.0	7801.6 x 6338.8	Two front-region PicoCal surfaces, entry and exit, as a simple volume proxy.
MUON	4	4	15270.0 to 18870.0	12119.2 x 10057.3	Muon stations M2-M5, rectangular acceptances with central beam holes.

Configuration

{"ft": {"stations": {"T1": 8, "T2": 8, "T3": 10}}, "include": ["TV", "RICH1", "UP", "MP", "FT", "RICH2", "PicoCal", "MUON"], "name": "lhcb_lower_up_gdml_ft_8_8_10", "source": "ViewLHCb lower ROOT/GDML extracted surfaces", "tv": {"grouping": "paired_

TV XY surfaces

50 planes, z = -287.500 to 557.650 mm, XY envelope = 90.3 x 92.0 mm



x-y view; holes are shown by white cutouts where present

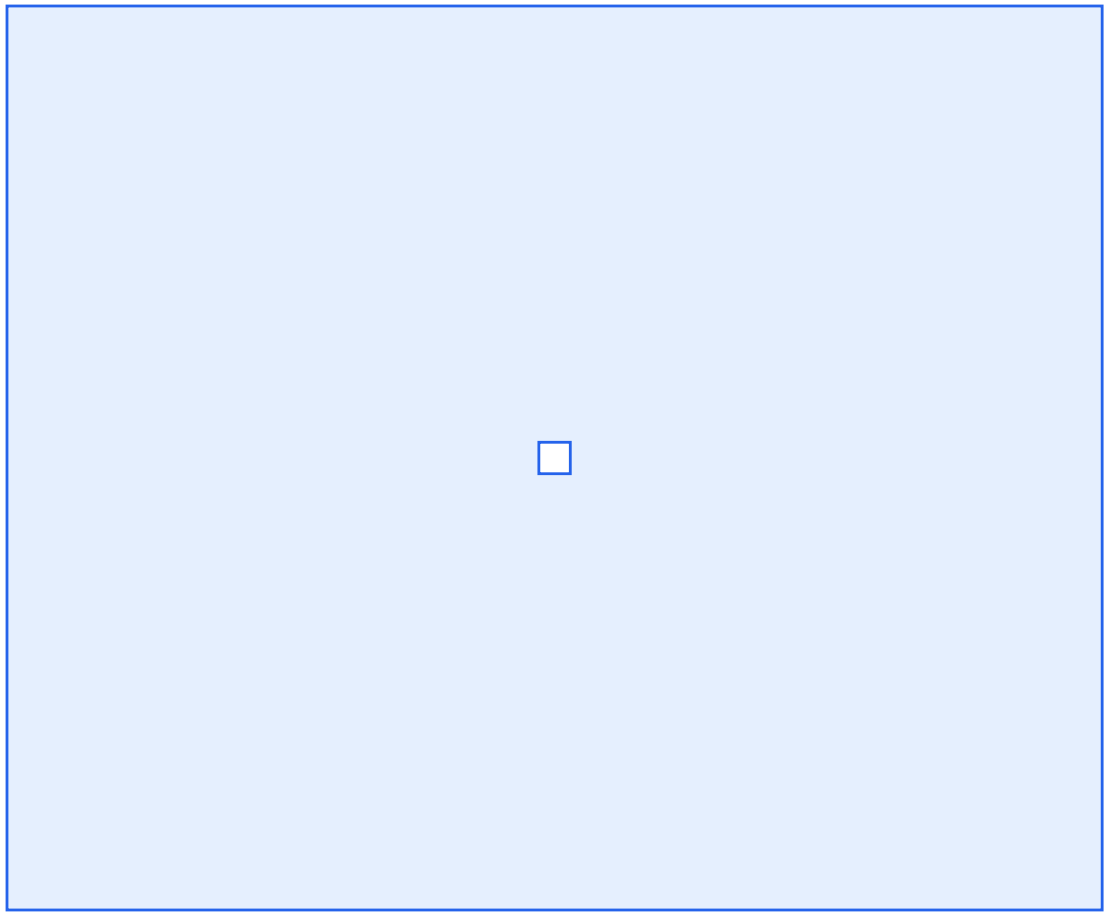


Layers

TV_layer1: z=-287.500
TV_layer2: z=-275.000
TV_layer3: z=-262.500
TV_layer4: z=-250.000
TV_layer5: z=-237.500
TV_layer6: z=-225.000
TV_layer7: z=-212.500
TV_layer8: z=-200.000
TV_layer9: z=-167.500
TV_layer10: z=-155.000
TV_layer11: z=-127.500
TV_layer12: z=-115.000
TV_layer13: z=-94.750
TV_layer14: z=-82.250
TV_layer15: z=-73.150
TV_layer16: z=-60.650
... plus 34 more

RICH1 XY surfaces

2 planes, z = 795.000 to 2237.200 mm, XY envelope = 1384.1 x 1142.5 mm



x-y view; holes are shown by white cutouts where present

z placement of all planes

795.0 mm 2237.2 mm

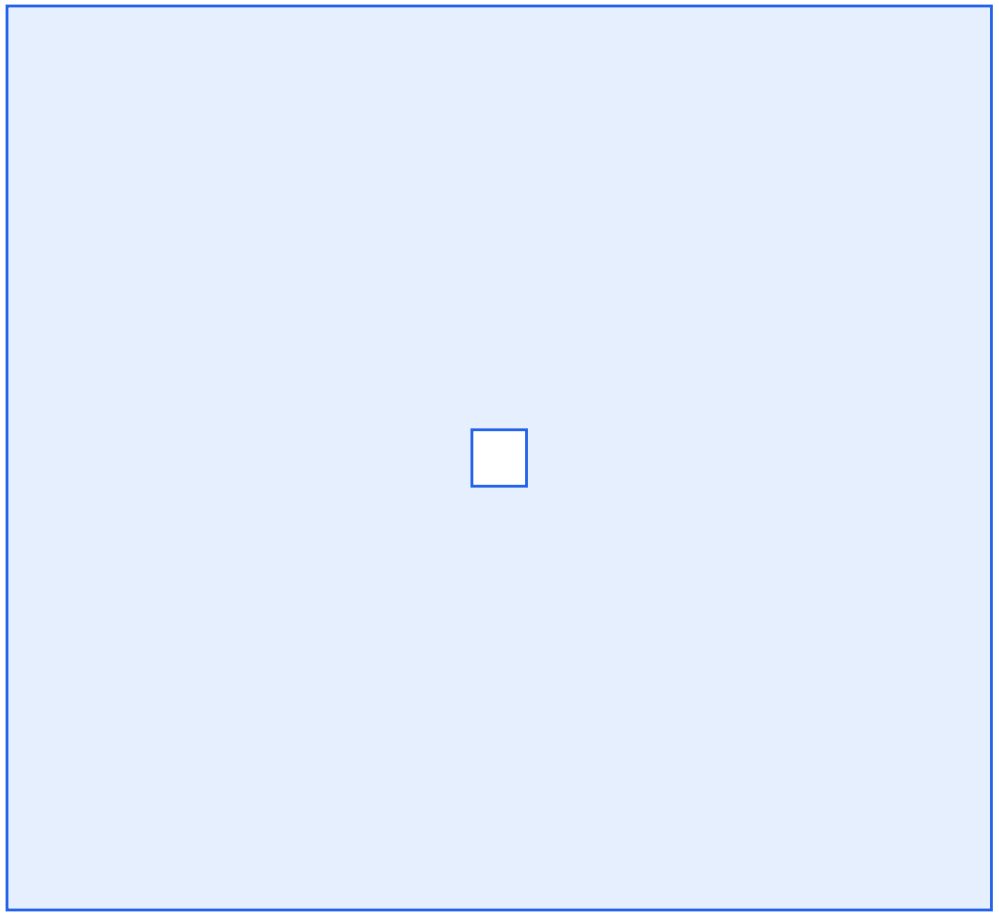
Layers

rich1_layerEntry: z=795.000

rich1_layerExit: z=2237.200

UP XY surfaces

16 planes, z = 2314.031 to 2655.969 mm, XY envelope = 1393.7 x 1279.8 mm



x-y view; holes are shown by white cutouts where present

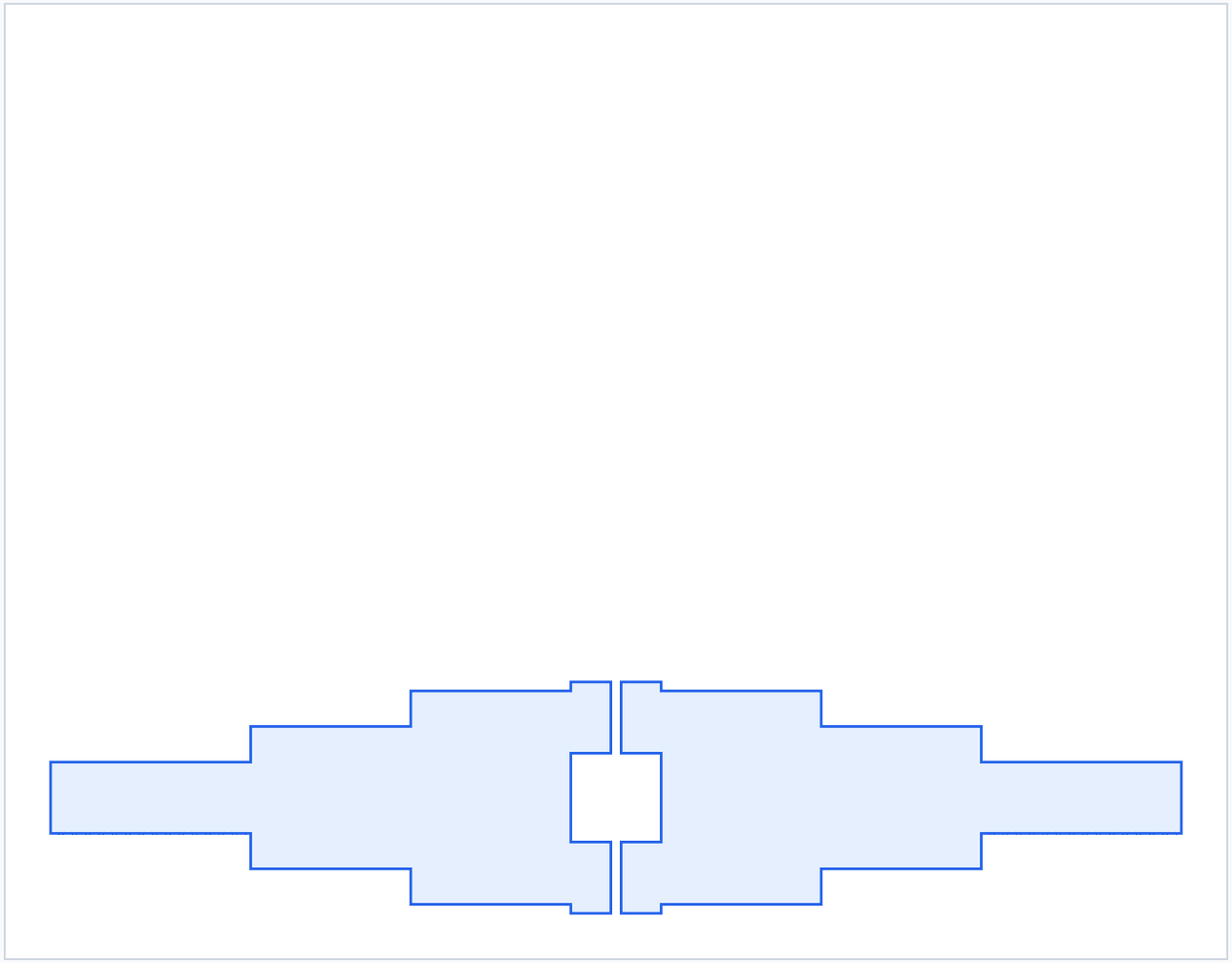


Layers

UP_layer1_surface1: z=2314.031
UP_layer1_surface2: z=2314.232
UP_layer1_surface3: z=2320.768
UP_layer1_surface4: z=2320.969
UP_layer2_surface1: z=2369.031
UP_layer2_surface2: z=2369.232
UP_layer2_surface3: z=2375.768
UP_layer2_surface4: z=2375.969
UP_layer3_surface1: z=2594.031
UP_layer3_surface2: z=2594.232
UP_layer3_surface3: z=2600.768
UP_layer3_surface4: z=2600.969
UP_layer4_surface1: z=2649.031
UP_layer4_surface2: z=2649.232
UP_layer4_surface3: z=2655.768
UP_layer4_surface4: z=2655.969

MP XY surfaces

5 planes, z = 7665.500 to 9132.500 mm, XY envelope = 3288.4 x 672.6 mm



x-y view; holes are shown by white cutouts where present



Layers

MP_layer1: z=7665.500

MP_layer2: z=7825.500

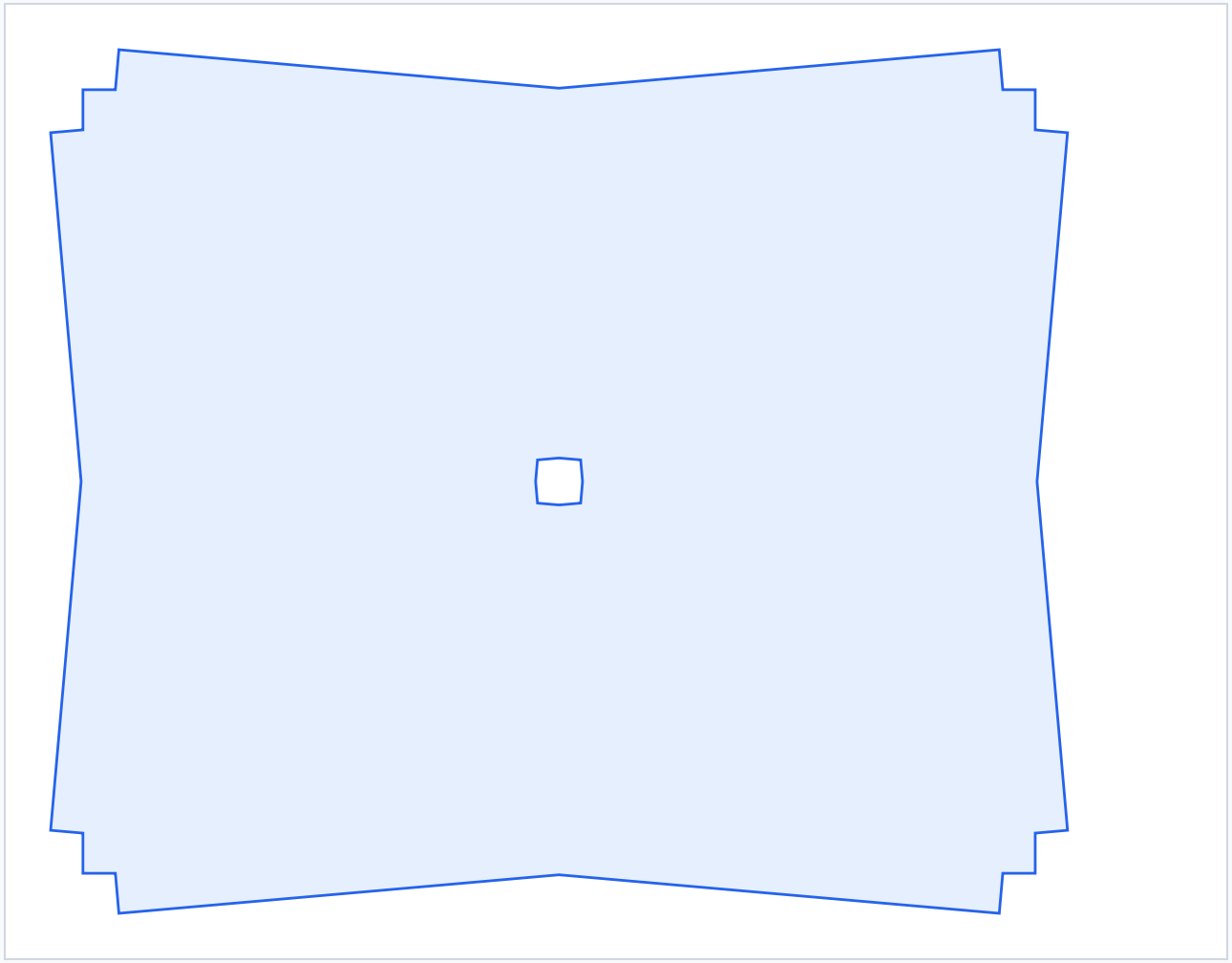
MP_layer3: z=8380.500

MP_layer4: z=8972.500

MP_layer5: z=9132.500

FT XY surfaces

12 planes, z = 7902.700 to 9384.700 mm, XY envelope = 5652.3 x 4798.7 mm



x-y view; holes are shown by white cutouts where present

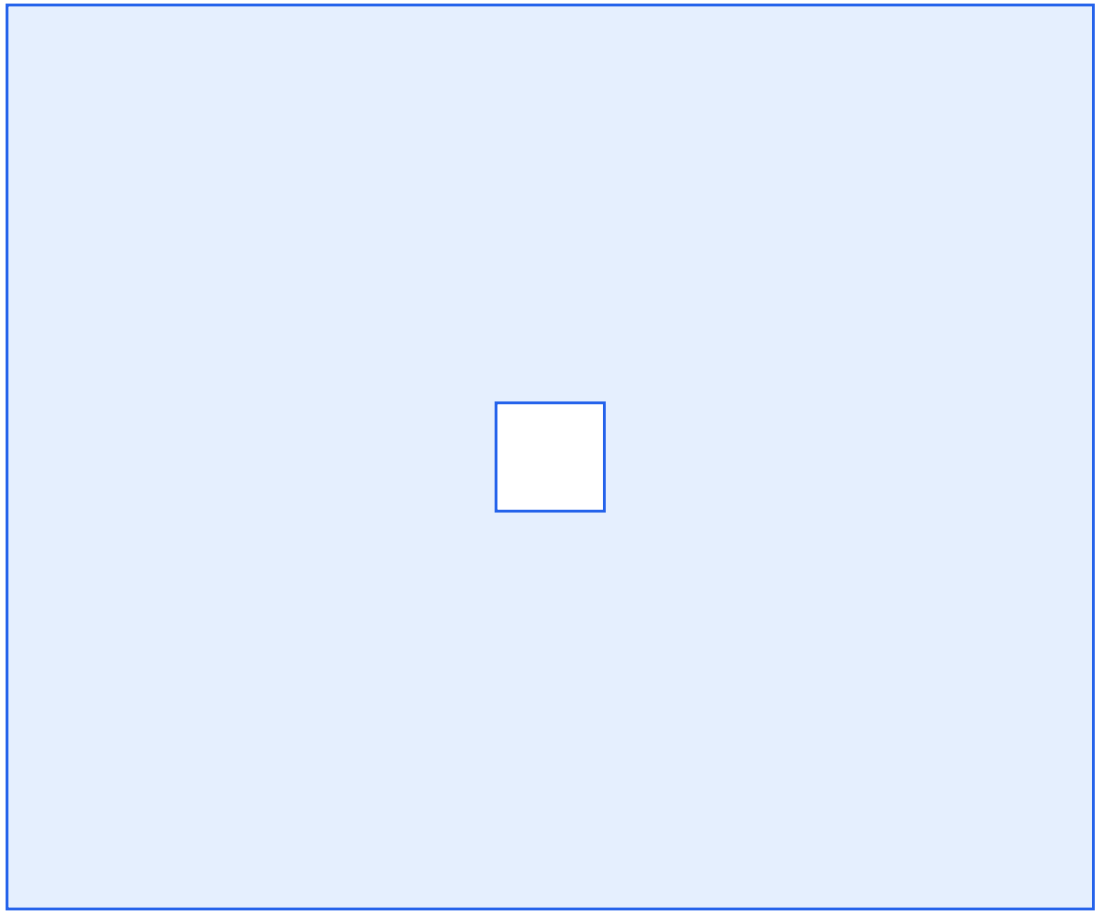


Layers

scifi_layer1_T1_X1: z=7902.700
scifi_layer2_T1_U: z=7947.699
scifi_layer3_T1_V: z=8032.701
scifi_layer4_T1_X2: z=8077.700
scifi_layer5_T2_X1: z=8524.700
scifi_layer6_T2_U: z=8569.699
scifi_layer7_T2_V: z=8654.701
scifi_layer8_T2_X2: z=8699.700
scifi_layer9_T3_X1: z=9209.700
scifi_layer10_T3_U: z=9254.699
scifi_layer11_T3_V: z=9339.701
scifi_layer12_T3_X2: z=9384.700

RICH2 XY surfaces

2 planes, z = 9487.500 to 11839.200 mm, XY envelope = 2855.1 x 2375.8 mm



x-y view; holes are shown by white cutouts where present

z placement of all planes

9487.5 mm 11839.2 mm

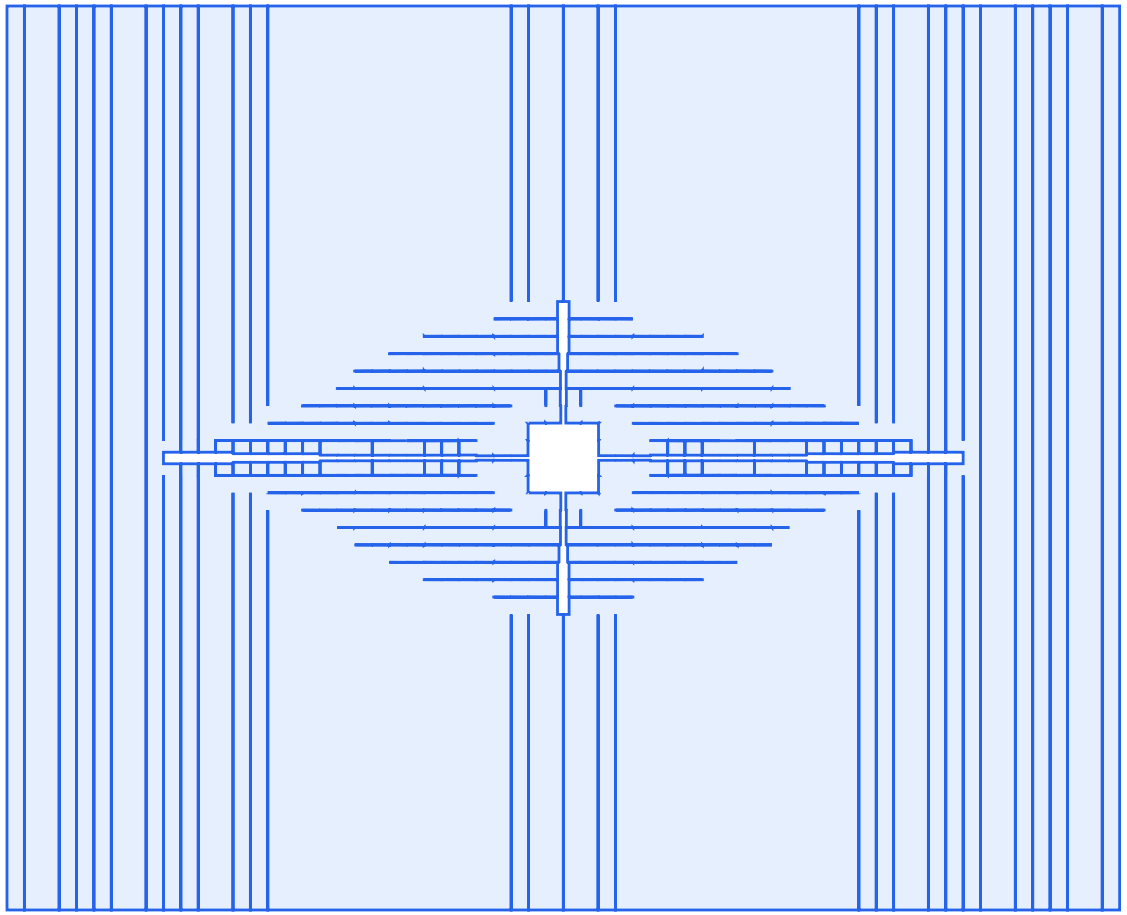
Layers

rich2_layerEntry: z=9487.500

rich2_layerExit: z=11839.200

PicoCal XY surfaces

2 planes, z = 12565.869 to 12601.012 mm, XY envelope = 7801.6 x 6338.8 mm



x-y view; holes are shown by white cutouts where present

z placement of all planes

12565.9 mm — 12601.0 mm

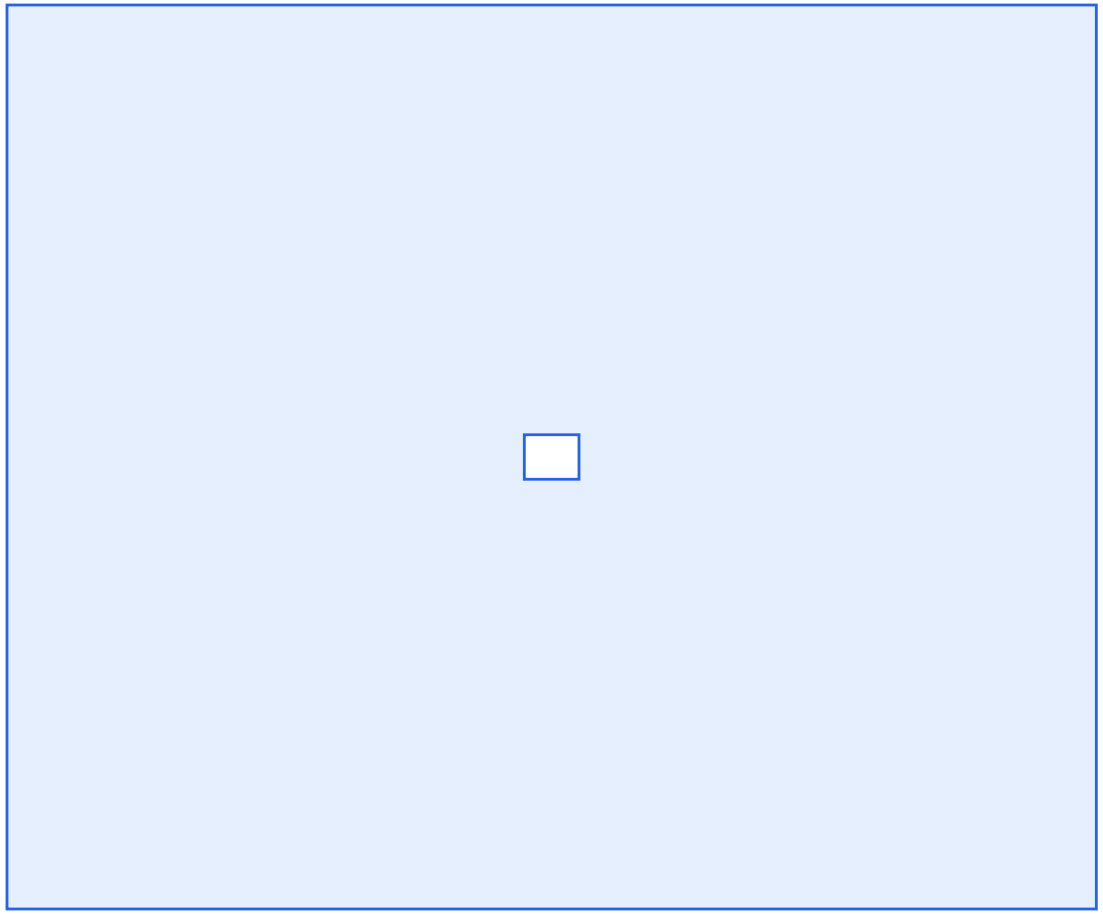
Layers

picocal_layerEntry: z=12565.869

picocal_layerExit: z=12601.012

MUON XY surfaces

4 planes, z = 15270.000 to 18870.000 mm, XY envelope = 12119.2 x 10057.3 mm



x-y view; holes are shown by white cutouts where present



Layers

muon_m2: z=15270.000

muon_m3: z=16470.000

muon_m4: z=17670.000

muon_m5: z=18870.000

Available UP and FT XY variants

Use this page to compare choices before building the final setup.

gdml



1393.7 x 1279.8 mm

y8



1393.7 x 964.9 mm

y8x6



1164.7 x 964.9 mm

FT 8 X



4234.4 x 4354.0 mm

FT 8 U



4597.8 x 4706.5 mm

FT 8 V



4597.8 x 4706.5 mm

FT 10 X



5293.0 x 4354.0 mm

FT 10 U



5652.3 x 4798.7 mm

FT 10 V



5652.3 x 4798.7 mm

FT 12 X



6351.6 x 4354.0 mm

FT 12 U



6706.9 x 4891.0 mm

FT 12 V



6706.9 x 4891.0 mm